

Expense Tracker

Business Requirements Document · Version 1.0

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Prepared for	Platform Engineering
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1. Purpose of this document

This document states what the Expense Tracker service must do, in business terms. It is the input to the API design, and it deliberately does not describe endpoints, HTTP methods, resource paths, payload structures or status codes — those are design decisions to be made by the engineering team from the requirements below.

Where this document names a piece of information the service must hold, it uses the business term for it. Where it states a rule, that rule is expected to hold true no matter how the API is eventually shaped.

2. Background and business problem

Northbeam Consulting has around 400 consultants, most of whom travel to client sites and incur expenses weekly — travel, accommodation, meals, software subscriptions and client entertainment. Expenses are currently submitted as spreadsheets emailed to Finance with photographed receipts attached.

This process is failing in four measurable ways:

- Reimbursement takes an average of 24 days from submission to payment, against a published commitment of 10 working days.
- Roughly 12% of submissions are returned to the employee at least once, most often for a missing receipt or an expense outside the claim window.
- Finance cannot report on spend by category or client until a quarter has closed, so budget overruns are found after the fact.
- There is no reliable audit trail of who approved what, which was raised as a finding in the last external audit.

The business has approved building an Expense Tracker service to replace the spreadsheet process. Several client applications will consume it — a mobile app for consultants, a web console for approvers, and a scheduled export into the payroll system — so the service must be usable by all three rather than built for any one of them.

3. Scope

In scope

- Recording, amending and deleting individual expenses.
- A centrally maintained list of expense categories.
- Attaching receipt images to an expense.
- Grouping expenses into a claim and submitting it for approval.

- Approving or rejecting a claim, with a recorded decision.
- Marking an approved claim as reimbursed.
- Retrieving expenses and claims for reporting.

Out of scope for this release

- Corporate card feeds and automatic transaction matching.
- Optical receipt reading or automatic data extraction from images.
- Mileage calculation from mapping data.
- Direct payment execution — the service records that a claim was reimbursed; payment itself remains with payroll.
- Per-client or per-project budget enforcement.

4. Who uses the service

ROLE	WHAT THEY DO
Employee	Records expenses, attaches receipts, groups them into a claim and submits it. Can see only their own expenses and claims.
Approver	Usually the employee's engagement manager. Reviews submitted claims and either approves or rejects them. Can see claims submitted to them.
Finance	Maintains the category list, marks approved claims as reimbursed, and retrieves expense data for reporting. Can see all expenses and claims.

5. Information the service must hold

Described in business terms. Naming, grouping and structure are design decisions.

An expense

INFORMATION	MEANING
Amount and currency	The amount as it appeared on the receipt, in the currency it was paid in.
Amount in EUR	The same amount converted to euros, the company reporting currency.
Date incurred	The date the money was spent, which is not the date it was recorded.
Category	Exactly one category, from the list Finance maintains.
Description	A short free-text explanation. Required.
Payment method	Whether the employee paid personally or used a company card.
Client reference	Optional. The client engagement the expense should be billed to.
Receipts	Zero or more attached receipt images.
Owner	The employee the expense belongs to.
Recorded at	When the expense was first entered.

A category

INFORMATION	MEANING
Name	For example Travel, Accommodation, Meals, Software, Client Entertainment.
Active or retired	Retired categories must remain readable on historic expenses but must not be selectable on new ones.
Receipt threshold	The amount above which a receipt becomes mandatory for this category. Some categories require one at any amount.

A claim

INFORMATION	MEANING
Title	Set by the employee, for example "March 2026 — Client site visits".
Expenses	One or more expenses. An expense may belong to at most one claim.
Total	The sum of the expenses it contains, in EUR.
Status	Where the claim has reached. See section 7.
Submitted at	When the employee submitted it.
Decision	Who approved or rejected it, when, and — if rejected — why.

6. Functional requirements

Recording expenses

ID	REQUIREMENT	PRIORITY
BR-01	An employee must be able to record an expense, supplying at minimum an amount, a currency, the date it was incurred, a category and a description.	Must
BR-02	An employee must be able to amend or delete an expense while it is not part of a submitted claim.	Must
BR-03	An employee must be able to retrieve their own expenses, and to narrow that list by date range, category and whether the expense has been claimed yet.	Must
BR-04	An employee must be able to attach one or more receipt images to an expense, and to remove an attachment they added in error.	Must
BR-05	An employee must not be able to see, amend or delete an expense belonging to another employee.	Must

Categories

ID	REQUIREMENT	PRIORITY
BR-06	Anyone recording an expense must be able to retrieve the list of categories currently available to them.	Must
BR-07	Finance must be able to add a category, rename one, change its receipt threshold, and retire one.	Must
BR-08	Only Finance may change the category list. Employees and approvers may read it but not modify it.	Must

Claims and approval

ID	REQUIREMENT	PRIORITY
BR-09	An employee must be able to group their unclaimed expenses into a claim, give it a title, and submit it for approval.	Must
BR-10	An employee must be able to see the current status of each of their claims, and the reason for any rejection.	Must
BR-11	An approver must be able to retrieve the claims awaiting their decision, including every expense and receipt within them.	Must
BR-12	An approver must be able to approve a claim, or reject it with a reason.	Must
BR-13	Finance must be able to mark an approved claim as reimbursed once payroll has paid it.	Must
BR-14	Finance must be able to retrieve all expenses across the company for a given period, with the category and client reference on each, for reporting.	Must
BR-15	An approver should be able to return a claim for correction without formally rejecting it.	Should

7. Business rules

These rules must hold regardless of how the API is designed, and are the basis on which the service will be accepted.

ID	REQUIREMENT	PRIORITY
BR-16	An expense amount must be greater than zero. Refunds and credits are not handled in this release.	Must
BR-17	The date an expense was incurred must not be in the future.	Must
BR-18	An expense incurred more than 90 days ago cannot be added to a claim. The expense may still be recorded and read.	Must
BR-19	A receipt is mandatory where the expense amount exceeds the receipt threshold for its category. A claim containing such an expense without a receipt cannot be submitted.	Must
BR-20	Client Entertainment requires a receipt at any amount, and requires a client reference.	Must
BR-21	A claim must contain at least one expense. An empty claim cannot be submitted.	Must
BR-22	An expense may belong to at most one claim at a time.	Must
BR-23	A claim moves through the following states only: draft, submitted, approved, rejected, reimbursed. A claim may be submitted only from draft; approved or rejected only from submitted; reimbursed only from approved.	Must
BR-24	A rejected claim returns to draft so the employee can correct and resubmit it. Its rejection reason must be preserved.	Must
BR-25	Expenses inside a submitted, approved or reimbursed claim cannot be amended or deleted.	Must
BR-26	An approver cannot approve a claim they submitted themselves.	Must
BR-27	A reimbursed claim is final. It cannot be reopened, amended or deleted.	Must

ID	REQUIREMENT	PRIORITY
BR-28	Where an expense is recorded in a currency other than EUR, both the original amount with its currency and the EUR equivalent must be stored. The conversion rate applied is the rate on the date the expense was incurred.	Must
BR-29	Every change of claim status must record who made it and when, and that record must not be alterable afterwards.	Must
BR-30	Retiring a category must not alter or invalidate expenses already recorded against it.	Must

8. Non-functional requirements

ID	REQUIREMENT	PRIORITY
NFR-01	Every caller must be authenticated. No part of the service may be reachable anonymously.	Must
NFR-02	Callers must only be able to reach the information their role permits, as set out in section 4. An attempt to exceed that must be refused, not silently filtered.	Must
NFR-03	Retrieving a list of expenses must respond within 500 ms for a month of data for one employee.	Must
NFR-04	The service must support 500 concurrent users, with a peak at month end of roughly four times the daily average.	Must
NFR-05	Lists that can grow without bound must be returned in pages, so a client is never obliged to load a whole year at once.	Must
NFR-06	When a request is refused, the response must explain which rule was broken clearly enough for a client application to show the employee a useful message.	Must
NFR-07	Receipt images may be up to 10 MB each, in JPEG, PNG or PDF format. A maximum of 10 receipts may be attached to one expense.	Must
NFR-08	The service must be documented well enough that a client team can integrate without asking the owning team questions.	Must
NFR-09	All monetary amounts must be exact. Rounding errors in reimbursement totals are not acceptable.	Must

9. How success will be judged

- A consultant can record a week of expenses, attach receipts and submit a claim without contacting Finance for help.
- A claim that breaks any rule in section 7 is refused, with a reason the employee can act on.
- Median time from submission to reimbursed falls below 10 working days within one quarter of launch.
- Finance can produce a spend-by-category report for any period without waiting for the quarter to close.
- Every status change on every claim can be traced to a person and a timestamp.

10. Open questions

These are unresolved and should be raised during design rather than assumed.

- Should the 90-day claim window be waivable by Finance in exceptional cases?

- Where should currency conversion rates come from, and who owns that feed?
- Does a claim above a certain value need a second approver?
- How long must receipt images be retained after reimbursement?

End of document. Section 1 applies throughout: this states what the service must do, not how the API should be shaped.